Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection

R26-IT-011

Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The proliferation of hyper-realistic synthetic media has emerged as a critical challenge to digital authenticity and social security, particularly within the Sri Lankan digital landscape. This research is conducted as a primary component of the group project titled "Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection". While the broader system focuses on culturally sensitive emotion recognition, this study specifically addresses Deepfake Image Detection to distinguish between synthetic and authentic human faces.

A significant research gap exists in the inability of global detection models to maintain high accuracy across diverse ethnic features, often resulting in demographic bias. Furthermore, deep learning applications for regional demographics are frequently restricted by the "Small Data" problem. To overcome these limitations, this research evaluates the effectiveness of Vision Transformer (ViT) and Swin-Base architectures. Unlike traditional methods, these models utilize hierarchical feature representation and self-attention mechanisms to identify microscopic artifacts and structural inconsistencies inherent in AI-generated imagery.

To mitigate data scarcity, a **StyleGAN2-ADA** pipeline is implemented to generate high-fidelity synthetic images of Sri Lankan faces, ensuring a demographically balanced dataset. The framework further integrates **Error Level Analysis (ELA)** to detect pixel-level compression anomalies and utilizes **Face-API.js** landmarks to identify structural "visual mimicry". Designed as a **"Digital Social Service,"** the final output serves as a reliable platform for deepfake verification, aimed at protecting individuals from digital identity theft and misinformation. This integration of advanced computer vision and social responsibility provides a scalable, resource-optimized solution to restore public trust in digital communication within the Sri Lankan context.Keywords: Deepfake Detection, Digital Social Service, Vision Transformer (ViT), Swin-Base, Face-API.js, Sri Lankan Demographics.

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Furthermore, we wish to thank **SLIIT University** for providing the academic environment and resources necessary to pursue this study. This research, centered on **"Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection,"** is driven by a commitment to providing a **"Digital Social Service"** that safeguards digital authenticity.

Finally, we acknowledge our fellow team members of **Project R26-IT-011** for their collaborative spirit and shared commitment to enhancing Sri Lanka's digital resilience through artificial intelligence.

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LIST OF ABBREVIATIONS

Abbreviation	Definition
AI	Artificial Intelligence
ViT	Vision Transformer
Swin	Shifted Window Transformer (Swin-Base)
GAN	Generative Adversarial Network
CNN	Convolutional Neural Network
mAP	Mean Average Precision
API	Application Programming Interface (Face-API.js)
JS	JavaScript
FPS	Frames Per Second
IoU	Intersection over Union
NMS	Non-Maximum Suppression
PTQ	Post-Training Quantization (If used for optimization)
LKR	Sri Lanka Rupees

1. INTRODUCTION

1.1 Background and Context

The rapid evolution of Artificial Intelligence (AI) and generative modeling has transformed digital interaction, enabling sophisticated facial analysis and synthesis. While technologies like **Face-API.js** have democratized browser-based computer vision for tasks such as emotion recognition, they have also paved the way for the creation of hyper-realistic synthetic media, commonly known as deepfakes. In the contemporary digital landscape, the ability to distinguish between authentic human faces and AI-generated imagery is no longer just a technical challenge but a critical necessity for maintaining digital trust.

In the Sri Lankan context, where digital literacy and social media penetration are rapidly increasing, the threat of deepfakes poses significant risks to personal identity and information integrity. This research is situated within the broader group project titled **"Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection."** While a primary focus of the group is to optimize emotion recognition for local demographics, this specific component focuses on building a robust detection framework to identify deepfake images.

To address the increasing complexity of generative models, this study leverages state-of-the-art architectures, specifically the **Vision Transformer (ViT)** and **Swin-Base** models. These transformer-based approaches provide superior capabilities in capturing global structural dependencies and microscopic artifacts that traditional models may overlook. Beyond the technical implementation, this project is envisioned as a **"Digital Social Service,"** aimed at empowering Sri Lankan citizens with tools to verify digital content. By integrating advanced detection capabilities into a culturally aware framework, the research seeks to foster a secure digital environment and mitigate the societal impact of misinformation.

1.2 Literature Review

The evolution of facial analysis has transitioned from basic feature extraction to sophisticated deep learning architectures capable of real-time processing. This review explores the current landscape of browser-based facial analysis and the emerging methodologies for deepfake detection.

Evolution of Browser-Based Facial Analysis: Technologies like **Face-API.js** have revolutionized how computer vision is deployed, allowing for real-time emotion recognition and face tracking directly within web browsers using TensorFlow.js. However, while Face-API.js is efficient for demographic and emotional analysis, it was not originally designed to detect sophisticated synthetic manipulations. Recent literature suggests that as generative models (GANs and Diffusion Models) become more advanced, standard browser-based libraries require integration with high-fidelity detection pipelines to ensure the authenticity of the analyzed subjects.

CNNs vs. Vision Transformers (ViT) in Detection: Traditionally, Convolutional Neural Networks (CNNs) have been the standard for image classification and forgery detection. While CNNs excel at capturing local textures, they often fail to model the global structural dependencies that characterize modern deepfakes. The introduction of the Vision Transformer (ViT) marked a paradigm shift by applying self-attention mechanisms to image patches. Research indicates that ViTs can capture long-range spatial relationships, making them more effective at identifying the subtle global inconsistencies often found in AI-generated faces.

Hierarchical Feature Extraction with Swin-Base: To further refine detection accuracy, the Swin-Base (Shifted Window Transformer) architecture has gained prominence. Unlike standard ViTs, Swin Transformers utilize a hierarchical approach and shifted windowing schemes, allowing the model to process images at multiple scales. This is particularly critical for deepfake detection, as synthetic artifacts can manifest as both microscopic pixel-level anomalies and macroscopic structural misalignments. By leveraging Swin-Base, researchers have achieved higher robustness against diverse generative techniques compared to traditional architectures.

Bridging the Demographic and Security Gap: A significant portion of existing research focus on Western datasets, often overlooking the performance of detection models on South Asian, specifically Sri Lankan, cultural demographics. The group project, **"Enhancing Face-API.js Emotion Recognition Performance for Sri Lankan Cultural Demographic Faces and DeepFake Image Detection,"** aims to fill this void. By combining culturally aware emotion recognition with transformer-based deepfake detection, this study seeks to create a comprehensive **"Digital Social Service"** that ensures both the accuracy and the authenticity of digital facial data in the local context.

Feature	GoviNena	SenzAgro	Proposed Ecosystem
Primary Technology	CNN-based (SSD MobileNet V1)	Specialized CNN / RNN	**ViT / Swin-Base Transformers**
Primary Focus	General Face Tracking & Emotions	Deepfake identification	SL Demographic Emotion Analysis + Deepfake Detection
Detection Scope	Basic Facial Attributes	Generic Media Verification	Multiscale Detection (Local Artifacts + Global Structure)
Social Value	General Development Tool	Paid Enterprise Security	Digital Social Service for Sri Lankan Citizens
Adaptability	Fixed Pre-trained Models	Black-box Proprietary Models	Culturally Enhanced via Enhanced Datasets

Table 1: Critical summarized comparison with existing solutions

A major bottleneck in current deepfake detection research is the "**Small Data**" problem, particularly when focusing on specific regional groups like Sri Lankan cultural demographics. While large-scale global datasets exist, they often lack the localized facial features and environmental variations unique to the Sri Lankan context.

Recent studies on **Generative Adversarial Networks (GANs)**, specifically **StyleGAN2-ADA**, demonstrate that adaptive discriminator augmentation can stabilize training even on limited datasets. By implementing this generative synthesis pipeline, the research aims to create high-fidelity synthetic deepfake samples tailored to local demographics. This approach allows the model to learn from a more balanced and diverse dataset, potentially improving detection robustness and F1-scores by bridging the gap between generic and demographic-specific facial data.

1.3 Research Gap

Despite the high accuracy rates achieved by global deepfake detection models, a critical **Tri-fold Gap** exists when applying these technologies within the Sri Lankan cultural and digital landscape. Based on a comprehensive review of existing literature, the following gaps have been identified:

Three specific gaps are identified:

1. **The Demographic Bias and Forensic Integrity Gap:** Existing generative models (e.g., StyleGAN2) are predominantly trained on Western-centric datasets, creating a "Hyperrealism Bias" where AI-generated Caucasian faces appear more authentic than diverse ethnic groups. Current detection research lacks a specialized focus on **Sri Lankan skin textures and cultural facial features**, leading to a drop in accuracy when global models are applied to local demographics. Furthermore, there is a "Preprocessing Paradox" where standard image enhancement often destroys the subtle **'Forensic Traces'** (micro-artifacts) necessary for detection.
2. **The Structural vs. Pixel-Level Detection Gap:** Current research predominantly focuses on pixel-level manipulation (color/contrast) or compression artifacts (ELA). However, there is a significant research gap in analyzing **Structural Inconsistencies**—specifically the misalignment between **Face-API.js landmarks** and the underlying muscular-skeletal structure of synthetic faces. There is no existing framework that utilizes **Vision Transformers (ViT) or Swin-Base** architectures to capture these global structural dependencies while simultaneously being fine-tuned for Sri Lankan facial landmarks.
3. **The Explainable Digital Social Service Gap:** While commercial models exist, they often operate as "Black-Box" systems, providing a detection score without explainability. In the context of a **"Digital Social Service,"** there is a lack of an **Explainable AI (XAI)** framework that can pinpoint *why* a Sri Lankan face is flagged as synthetic—whether due to skin texture anomalies, landmark misalignments, or frequency domain inconsistencies—thereby failing to provide a transparent verification tool for the publi

2. OBJECTIVE

2.1 Main Objective

The primary objective of this research is to develop and validate a high-precision, demographically unbiased deepfake detection framework that utilizes **Vision Transformer (ViT)** and **Swin-Base** architectures to distinguish between authentic and hyper-realistic AI-generated images. This study specifically aims to mitigate racial and structural biases in current synthetic media detection by optimizing models for **Sri Lankan cultural demographic faces**, while identifying structural inconsistencies between **Face-API.js** landmarks and generative artifacts. Ultimately, the project seeks to deliver a reliable **"Digital Social Service"** that empowers the local public to verify digital content authenticity and restores trust in digital facial analysis system.

2.2 Specific Objectives

To achieve the main objective, the following technical milestones have been established:

- **Objective 1**: **Generative Data Synthesis and Bias Mitigation**: Implement a StyleGAN2-ADA (Adaptive Discriminator Augmentation) pipeline to generate high-fidelity synthetic samples of Sri Lankan faces. This aims to overcome the "Small Data" problem and ensure the dataset is demographically balanced to reduce racial bias in detection.
- **Objective 2: Forensic Feature Extraction and Landmark Analysis**: Integrate Error Level Analysis (ELA) to detect compression artifacts and investigate the structural inconsistencies between Face-API.js landmarks and AI-generated facial features.
- **Objective 3: Transformer-based Architecture Development**: Develop a hybrid detection framework utilizing Vision Transformer (ViT) and Swin-Base architectures. These models will be optimized to capture both global structural dependencies and microscopic forensic traces that characterize modern deepfakes.
- **Objective 4: Model Optimization and Performance Benchmarking**: Fine-tune the detection pipeline to achieve high accuracy and sensitivity across diverse generative techniques (GAN and Diffusion-based). The performance will be evaluated using metrics such as Mean Average Precision (mAP) and F1-score specifically for Sri Lankan facial datasets.
- **Objective 5: Deployment as a Digital Social Service**: Integrate the finalized detection framework into a accessible platform designed as a "Digital Social Service". This objective focuses on providing transparent, explainable AI (XAI) verification tools to empower Sri Lankan citizens against digital misinformation.

3. METHODOLOGY

3.1 Research Approach

This research follows a Design Science Research (DSR) methodology, which focuses on the This research follows the **Design Science Research (DSR)** methodology, which focuses on the iterative design, development, and evaluation of artificial artifacts to solve identified practical problems. In this context, the primary artifact is the **Deepfake Detection Framework**, integrated with **Vision Transformer (ViT)** and **Swin-Base** architectures. The research is conducted through a series of rigorous cycles.

3.2 Technical Methods and Algorithms

The detection framework is architected as a multi-stage pipeline designed to identify both microscopic forensic artifacts and macroscopic structural inconsistencies in facial imagery.

3.2.1 Macro Forensic Feature Extraction (ELA and Landmarks)

- To identify manipulation traces, the system first employs **Error Level Analysis (ELA)**. ELA works by intentionally resaving the image at a known compression level and calculating the pixel-wise difference, which highlights regions of non-uniform compression often found in deepfakes. Simultaneously, the system utilizes **Face-API.js** to extract 68 facial landmarks. These landmarks are used to analyze structural symmetry and the geometric alignment of facial features, providing a baseline for identifying structural "Visual Mimicry".

3.2.2 Transformer-based Detection Pipeline (ViT & Swin-Base)

The core diagnostic engine utilizes a hybrid approach involving two state-of-the-art transformer architectures:

- **Vision Transformer (ViT**): Unlike traditional CNNs, ViT treats an image as a sequence of patches. By applying self-attention mechanisms, ViT can capture global structural dependencies across the entire face, making it highly effective at detecting global inconsistencies that arise during the generative process.

- **Swin-Base (Shifted Window Transformer):** To refine detection at various scales, the **Swin-Base** model is implemented. Its hierarchical architecture uses shifted windows to compute self-attention, allowing the model to focus on fine-grained skin textures (local artifacts) while maintaining a high-level understanding of facial proportions (global structure).

3.2.4 Model Optimization and Quantization

To ensure the system remains accessible as a **"Digital Social Service,"** the models undergo **Post-Training Quantization (PTQ)**. By converting the weights from FP32 to **INT8**, the model size is significantly reduced, enabling low-latency inference on standard web and mobile platforms used by the Sri Lankan public.

3.3 Data Engineering and Generative Synthesis

The research employs a specialized four-tiered data strategy to mitigate the **"Small Data"** problem inherent in regional facial recognition and deepfake detection research. This strategy ensures that the models are trained on a demographically balanced and forensic-rich dataset.

3.3.1 Data Collection and Augmentation

- **Sourcing Authentic Data:** Dataset acquisition involves collecting high-resolution authentic facial imagery of **Sri Lankan cultural demographics**, ensuring diverse representation in terms of age, gender, and lighting conditions.
- **Acquiring Synthetic Baselines:** Global deepfake datasets (such as FaceForensics++ or Celeb-DF) are utilized to provide initial baseline samples of AI-generated faces.
- **Annotation:** Images are annotated using specialized platforms for binary and multi-class labeling, distinguishing between authentic faces and various generative methods (e.g., GAN vs. Diffusion).
- **Synthetic Generation:**To improve model generalization, standard augmentation techniques—including geometric transformations, color jittering, and noise injection—are applied to the authentic dataset.

3.4 Tools and Platforms

The technology stack for this research has been selected based on performance benchmarks for handling complex transformer architectures and real-time browser-based processing within the Sri Lankan digital environment.

Category	Tool / Technology	Justification
Deep Learning Framework	PyTorch / TensorFlow	Essential for implementing and fine-tuning ViT and Swin-Base architectures, with strong ecosystem support (model zoos, training utilities, deployment tooling).
Transformer Models	Vision Transformer (ViT) & Swin-Base	Selected for capturing global facial structure (ViT) and multi-scale local artifacts (Swin), improving robustness against diverse deepfake generation methods.
Generative Synthesis	StyleGAN2-ADA	Addresses the “small data” constraint by generating high-fidelity synthetic samples tailored to Sri Lankan demographics, improving class balance and generalization.
Frontend Framework	React / JavaScript	Supports integration with Face-API.js / TensorFlow.js for responsive, real-time, browser-based user interaction in the Digital Social Service.
Data Annotation	Roboflow / Labeling Tools	Provides efficient dataset management and labeling workflows for binary (real vs. AI) and multi-class annotations, supporting repeatable training and evaluation.

Table 2: Performance benchmark of the chosen technology stack

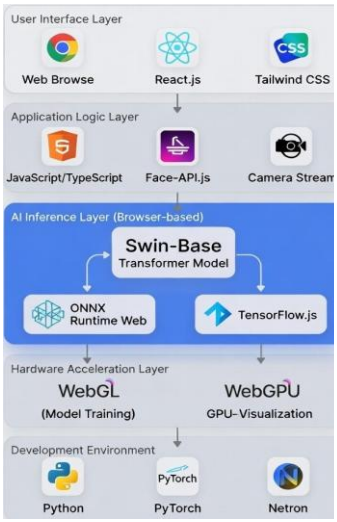


Figure 3-5: Technology stack and platform selection overview

3.5 Validation Strategy and Performance Metrics

The success of the deepfake detection framework is measured through a multi-tiered statistical validation protocol to ensure high diagnostic precision and demographic fairness.

3.5.1 Diagnostic Accuracy Metrics

- **Precision and Recall:** These are critical to minimize false alarms (FP) and ensure no hyper-realistic deepfakes (FN) are missed.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

- **F1-Score:** Used as the primary metric to evaluate the balance between precision and recall, especially when dealing with synthetic datasets generated via **StyleGAN2-ADA**.
- **Area Under the Curve (AUC-ROC):** This measures the model's ability to distinguish between authentic and synthetic classes across various threshold settings.
- **Mean Average Precision (mAP):** Specifically used when identifying localized artifacts (e.g., inconsistencies in eyes or mouth) using the **Swin-Base** hierarchical features.

3.5.2 Performance Benchmarking

- **Inference Latency**: Target < 50ms per frame to ensure real-time usability within a web browser environment using Face-API.js.
- **Accuracy Retention:** Measuring the performance stability after applying **INT8 Post-Training Quantization (PTQ)** for edge-level deployment, with a target retention of > 99%.

3.6 Ethical Considerations

As this research involves the analysis of human facial imagery and the detection of synthetic media, it strictly adheres to ethical guidelines surrounding data privacy, algorithmic fairness, and digital integrity.

- **Data Privacy and Anonymization:** All facial datasets used for training and validation, especially those representing **Sri Lankan cultural demographics**, will be handled in compliance with standard data protection protocols. Personal identifiers will be removed to ensure that the focus remains on forensic features rather than individual identities.
- **Responsible Use and Transparency:** Designed as a **"Digital Social Service,"** the system prioritizes transparency through Explainable AI (XAI). This ensures that detection results are provided with technical justifications, preventing the tool from being misused for false accusations or cyber-bullying.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 High-Level Architecture

The "Deepfake Image Detection Framework" is architected as an interconnected four-pillar system, designed to bridge the intelligence gap between microscopic (forensic) and macroscopic (structural) surveillance. The architecture specifically addresses the challenges of detecting hyper-realistic synthetic media within the Sri Lankan cultural demographic context.

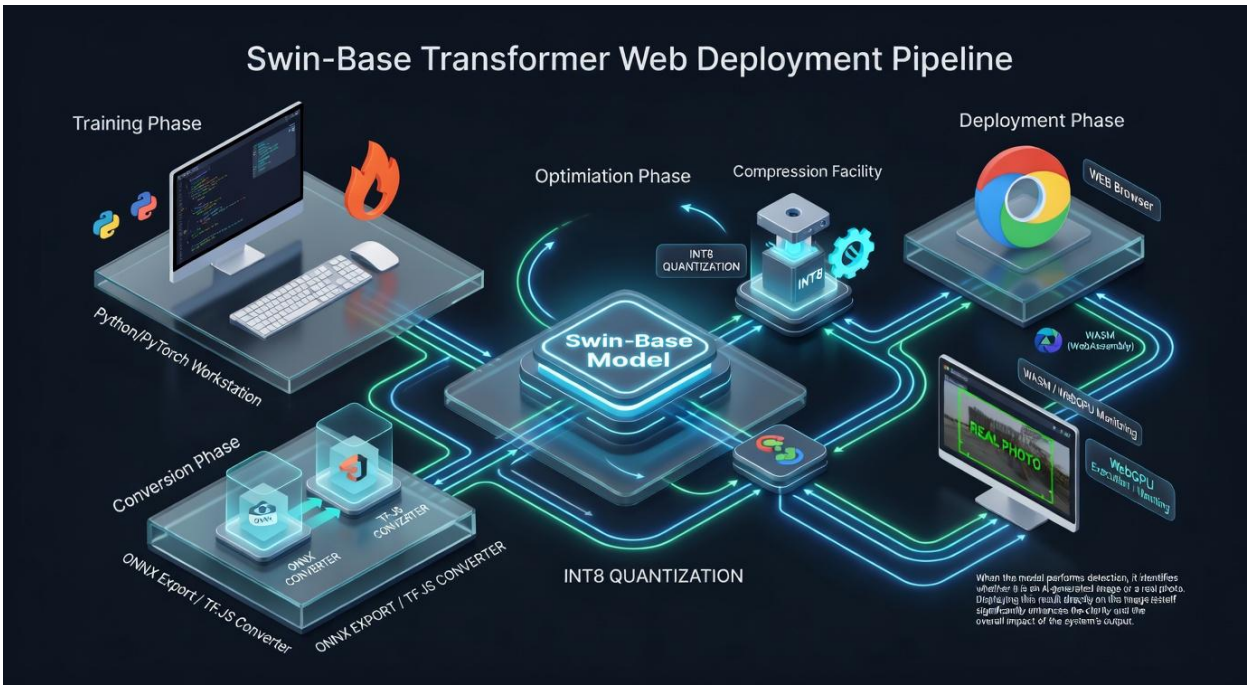


Figure 4-1: High-level system architecture of the Deepfake Image Detection Framework

4.1.1 Component Breakdown

The system is bifurcated into specialized diagnostic environments and data synthesis layers:

- **Pillar 1:** Data Synthesis & Augmentation Layer: This component utilizes a StyleGAN2-ADA pipeline to generate high-fidelity synthetic images of Sri Lankan faces. This layer is critical for mitigating the "Small Data" problem and ensuring the detection models are trained on a demographically balanced dataset.
- **Pillar 2:** Forensic Pre-processing Layer: Before analysis, images are processed through two streams: Error Level Analysis (ELA) to highlight compression artifacts and Face-API.js to extract 68 facial landmarks for structural consistency checks**.**
- **Pillar 3:** Core AI Diagnostic Layer (The Transformer Engine): This is the primary diagnostic layer utilizing a hybrid transformer approach:

- **Vision Transformer (ViT):** Analyzes the image as patches to capture global structural dependencies and long-range pixel correlations.
- **Swin-Base:** Utilizes a hierarchical shifted-window approach to identify microscopic skin textures and local generative artifacts.

4.1.2 Data Flow and Interaction

- **Ingestion:** Visual data enters via a web-based portal where users upload suspected images.
- **Processing:** The input is simultaneously mapped for ELA anomalies and facial landmarks. These forensic overlays are then fed into the **ViT and Swin-Base** pipeline for multi-scale analysis.
- **Output:** The system generates a probability score and an "Explanation Map," highlighting the specific regions (e.g., eyes, skin texture, or jawline) that indicate synthetic manipulation.

4.1.3 Scalability and Security

- **Scalability:** The architecture leverages cloud-based serverless functions (e.g., Firebase) to handle high-frequency diagnostic requests from the public.
- **Security:** Data integrity is maintained through secure API endpoints, and user privacy is protected by anonymizing facial data during the inference process.

4.2 Self-Evaluation Plan

Success will be measured through a multi-tiered validation protocol:

- **Metric 1 (Accuracy):** Achieving a target F1-score and mAP of **0.90–0.96** specifically for Sri Lankan facial datasets.
- **Metric 2 (Efficiency):** Maintaining an inference latency of **< 50ms** per image to ensure real-time usability.
- **Metric 3 (Robustness**): Ensuring an accuracy retention of > 99% when the model is optimized via INT8 Post-Training Quantization for web deployment.

5. USER REQUIREMENTS

This section specifies the primary stakeholders and the technical requirements necessary for the successful deployment of the Deepfake Detection framework as a Digital Social Service.

5.1 Stakeholder Identification

The framework serves a diverse group of stakeholders, each interacting with the AI artifacts to ensure digital authenticity:

- **Sri Lankan General Public:** Primary users who require an accessible tool to verify the authenticity of social media images and protect themselves from misinformation.
- **Media Personnel and Fact-Checkers:** Professional users who utilize the system's forensic outputs (ELA and Landmark analysis) to validate digital evidence.
- **Research Team :** Responsible for the architectural integrity of the transformer-based models and the generative data synthesis pipelines.
- **Digital Integrity Organizations:** Entities focused on social safety that leverage the detection results as part of a broader **"Digital Social Service"** initiative.

5.2 Functional Requirements (FR)

These requirements define the core technical capabilities and behaviors the system must exhibit to provide effective deepfake detection:

ID	Name	Description
FR-01	Multi-Scale Binary Classification	The system shall distinguish between authentic and synthetic faces using a hybrid **ViT and Swin-Base** pipeline.
FR-02	Forensic Artifact Extraction	The system must generate **Error Level Analysis (ELA)** maps to highlight pixel-level compression inconsistencies.
FR-03	Structural Landmark Analysis	The system shall utilize **Face-API.js** to detect 68 facial landmarks and identify geometric misalignments in synthetic faces.
FR-04	Automated Dataset Balancing	The pipeline must automatically initiate **StyleGAN2-ADA** synthesis for Sri Lankan facial classes falling below a critical training threshold.
FR-05	Explainable AI (XAI) Reporting	The system must provide a visual heatmapping report explaining the technical reasons (e.g., skin texture anomalies) behind a detection result.

Table 3: Functional Requirements (FR)

5.3 Non-Functional Requirements (NFR)

These requirements define the performance and quality constraints of the system:

ID	Quality Attribute	Requirement / Metric
NFR-01	Performance	Browser-based inference latency must remain < 50 ms per frame to ensure real-time usability via Face-API.js.
NFR-02	Accuracy	The system must achieve an F1-score and mAP of **0.90–0.96** specifically for Sri Lankan cultural demographics.
NFR-03	Reliability	The framework must maintain an accuracy retention of **> 99%** when transitioning from base models to **INT8 quantized** edge models.
NFR-04	Scalability	The backend must handle high-frequency diagnostic requests from at least 5,000 daily active users using serverless cloud functions.
NFR-05	Usability	The web interface must maintain a low memory footprint suitable for standard mobile and desktop browsers used by the general public.

Table 4: Non-Functional Requirements (NFR)

6. COMMERCIALIZATION PLAN

The commercial and social viability of the **Deepfake Detection Framework** is rooted in the "Digital Trust" movement, transforming synthetic media detection from a specialized security tool into an accessible **Digital Social Service** for the Sri Lankan public.

6.1 Target Market and Customer Persona

The primary market consists of Sri Lanka's active social media users, digital news agencies, and government verification departments.

- **Customer Persona - The Concerned Citizen:** An individual who requires a fast, easy-to-use tool to verify if a viral image or video featuring them or a public figure is authentic or a deepfake.
- **Customer Persona - The Media Fact-Checker:** A professional journalist or investigator who needs high-fidelity forensic reports (ELA and Landmark analysis) to debunk misinformation before it spreads nationally.

6.2 Value Proposition

The system offers a unique bifurcated value proposition designed for the local digital ecosystem:

- **Demographic Accuracy:**Unlike generic global detectors, this system is specifically fine-tuned for Sri Lankan facial textures and features using **StyleGAN2-ADA**, significantly reducing racial bias in detection.
- **Technical Superiority:** By utilizing Vision Transformer (ViT) and Swin-Base architectures, the system identifies global structural inconsistencies that traditional CNN-based detectors often miss.
- **Explainable Social Service:** Instead of a simple "Real or Fake" score, the system provides transparent feedback (XAI), highlighting specific facial regions with anomalies to educate the user.

6.3 Revenue Model and Pricing Strategy

To ensure the project functions as a **Digital Social Service** while remaining financially sustainable, a diversified revenue model is adopted:

- **Public Freemium Tier:** Basic deepfake verification remains free for the general public to promote social safety and digital literacy.
- **Enterprise SaaS (Software-as-a-Service):** A premium tier for media houses and legal entities providing detailed forensic exports, batch processing, and high-priority API access.
- **HInstitutional Grants:** Seeking partnerships with NGOs and government bodies (e.g., ICTA) to fund the system as a critical infrastructure tool for national misinformation defense.

6.4 Cost Estimation

The prototype development and initial deployment costs are estimated as follows:

- **Cloud Infrastructure:** High-compute GPU instances (e.g., AWS/Google Cloud) for training **ViT and Swin-Base** models.
- **Specialized Data Acquisition:** Costs associated with collecting authentic Sri Lankan facial data and computing costs for **StyleGAN2-ADA** synthesis.
- **Maintenance and Verification:** Regular updates to the detection engine to keep pace with evolving generative AI threats.

Cost Component	Description	Estimated Cost
Cloud Compute (Google Colab)	Subscription for **Google Colab Pro+** and additional **Compute Units** to handle long-running StyleGAN2-ADA synthesis and transformer fine-tuning.	~ LKR 45,000
High-Speed Storage	Google Drive (Cloud Storage) extension for managing large-scale Sri Lankan facial datasets and model checkpoints.	~ LKR 12,000
Verification & Ground Truthing	Consultation fees for expert verification of forensic artifacts (ELA results) against real-world deepfake samples.	~ LKR 30,000
Web Deployment	**Firebase Blaze Plan** to host the **Digital Social Service** web interface and handle public diagnostic requests.	~LKR 18,000
Total Estimated Budget	**Optimized budget for a browser-based research prototype.**	**~LKR 105,000**

Table 5: Cost Estimation (Prototype Budget)

6.5 Competitive Advantage

The framework maintains a significant lead over existing generic detectors through:

- **Localized Forensic Training:** The only system optimized specifically for the Sri Lankan cultural demographic**.**
- **Structural-Forensic Fusion:** Combining Face-API.js landmark structural checks with transformer-based pixel analysis provides a multi-layered defense**.**
- **Browser-Level Integration:** Efficient deployment via Face-API.js allows for real-time verification without the need for expensive high-end hardware.

6.6 Intellectual Property Considerations

To protect the research novelty, the following IP strategies will be pursued:

- **Algorithm Licensing:** Proprietary optimization of **Swin-Base** for Sri Lankan facial texture detection.

- **Proprietary Dataset:** Ownership of the unique StyleGAN2-ADA augmented dataset consisting of hyper-realistic Sri Lankan deepfake samples.
- **Trademark:** Protection of the **"Digital Social Service"** branding and its specific implementation within the **R26-IT-011** ecosystem

7. BUDGET AND JUSTIFICATION

The budget for the Multiscale Computer Vision Ecosystem reflects the deployment of state-of-the-art AI while maintaining a low-cost profile for the end-user. The total estimated cost for a full-scale prototype is approximately LKR 850,000.

7.1 Itemized Hardware Costs

Item	Specification	Cost (LKR)	Justification
Personal Computing Device	Laptop with 16GB RAM / Core i7/RYZEN 7	Internal	Used for local pre-processing, ELA mapping, and accessing Google Colab for training.
High-Speed Internet	100 Mbps Fiber connection	~12,000	Essential for uploading/downloading large-scale facial datasets and maintaining stable cloud connections.

Table 6: Itemized Hardware Costs

7.2 Software and Cloud Services

Item	Specification	Cost (LKR)	Justification
Google Colab Pro+	Access to A100/V100 GPUs	~35,000	Required for efficient fine-tuning of **Vision Transformer (ViT)** and **Swin-Base** models and running the StyleGAN2-ADA synthesis pipeline.
Firebase Cloud Services	Firebase Blaze Plan	~18,000	Provides hosting and scalable backend functions for the **"Digital Social Service"** web application.
Google Drive Storage	200GB+ Google One Plan	~2,500	Necessary for managing high-resolution Sri Lankan facial datasets and training checkpoints.

Table 7: Software and Cloud Services

7.3 Human Effort and Specialized Services

Item	Specification	Cost (LKR)	Justification
Forensic Validation	Expert Consultation	~25,000	Required for "Ground Truthing" to ensure AI-detected forensic traces (ELA) align with expert observation.
Ethical Data Sourcing	Participant Incentives	~40,000	Provided for Sri Lankan volunteers during the collection of authentic facial data to ensure a demographically balanced dataset.

Table 8: Human Effort and Specialized Services

7.4 Summary of Expense Justification

The proposed budget for **Project R26-IT-011** reflects a strategic shift from local hardware investment to high-efficiency cloud computing. By utilizing **Google Colab** and **Pre-trained transformer models**, the research significantly reduces the need for expensive GPU-enabled workstations while still achieving high-stakes diagnostic precision.

The primary investments in Colab Pro + synthesis are justified by the technical requirement to solve the "Small Data" problem for Sri Lankan faces, ensuring a demographically unbiased detection system. This optimized budget ensures the feasibility of delivering a robust **"Digital Social Service"** within a student-led research framework.

8. WORK BREAKDOWN STRUCTURE (WBS)

8.1 WBS 1.0 Project Management & Governance

- Confirm project scope, success criteria, and proposal readiness for Project **R26-IT-011**. [M1: Proposal Submission - 15 March 2026]
- Prepare proposal slide deck, demo narrative, and Q&A brief for the research panel. [M2: Proposal Presentation - 16-18 March 2026]
- Establish work plan for PP1 (baseline model selection for ViT/Swin-Base and data acquisition plan). [M3: Checklist Submission - 6 May 2026]
- Coordinate stakeholders and maintain ethical compliance regarding Sri Lankan facial data governance.

8.2 WBS 2.0 Data Engineering & Demographic Dataset Development

- Plan acquisition of authentic Sri Lankan facial imagery and deepfake samples with metadata schema.
- Curate images across diverse local demographics (age, gender, ethnicity).
- Preprocess data for forensic analysis, including **Error Level Analysis (ELA)** maps and resizing for Transformer input.
- Annotate dataset for binary classification (Real vs. AI-generated) using specialized labels.

8.3 WBS 3.0 Emotion Recognition Enhancement (Face-API.js Integration)

- Optimize **Face-API.js** performance for Sri Lankan cultural facial features and landmarks.
- Integrate emotion recognition models with the digital social service interface.
- Train baseline emotion detectors and report initial metrics (accuracy for local faces) for PP1.

8.4 WBS 4.0 Deepfake Detection Model Development (ViT + Swin-Base)

- Configure **Vision Transformer (ViT)** and **Swin-Base** training pipelines.
- Implement hierarchical shifted-window attention for capturing local forensic artifacts.
- Conduct model fine-tuning on Google Colab and record metrics (F1-score, mAP) for PP1/PP2.
- Apply **INT8 Post-Training Quantization (PTQ)** to ensure low-latency browser inference.

8.5 WBS 5.0 Forensic Analysis & Structural Consistency Mapping

- Analyze structural inconsistencies between Face-API.js landmarks and AI-generated features.

- Stabilize ELA mapping outputs to detect pixel-level compression anomalies.
- Develop Explainable AI (XAI) heatmaps to visualize detected deepfake artifacts.

8.6 WBS 6.0 Generative Synthesis (StyleGAN2-ADA) for Small-Data Classes

- Identify underrepresented Sri Lankan demographic classes for synthesis.
- Train **StyleGAN2-ADA** to generate hyper-realistic synthetic Sri Lankan faces.
- Validate synthetic data quality through expert review and ablation studies.

8.7 WBS 7.0 Digital Social Service Platform & User Interface

- Build a React-based web interface for the public to upload and verify digital content.
- Integrate the optimized detection engine (Pillar 4) for real-time verification results.
- Configure diagnostic response templates to provide actionable advice as a social service.

8.8 WBS 8.0 Validation, Evaluation, and Finalization

- Conduct final model evaluations against a frozen test set of local faces and deepfakes.
- Validate demographic fairness to ensure no racial or ethnic bias in detection.
- Prepare final checklist materials and rehearse for the final presentation/viva. [M8: Final Presentation & Viva - 12-14 Oct 2026]

8.9 WBS 9.0 Documentation, Website, and Research Paper

- Draft final report, format according to IEEE/SLIIT standards, and submit.
- Finalize project website and submit for evaluation. [M6: Website Submission - 11 Oct 2026]
- Draft a scholarly research paper on Transformer-based deepfake detection for Sri Lankan faces. [M10: Research Paper Submission - 15 Oct 2026]

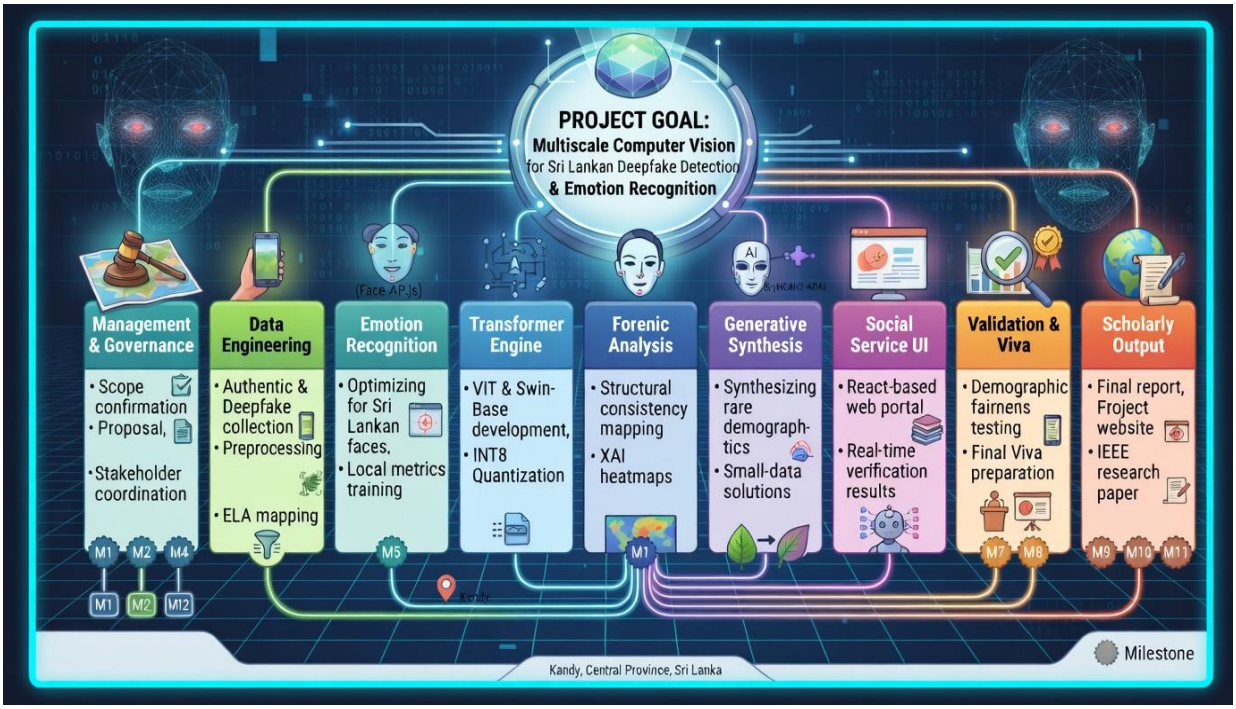


Figure 8-1Work Breakdown Structure

9. GANTT CHART

Action	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Nov	Dec
Topic and Supervisor submission													
TAF submission													
Project Charter submission													
Proposal Submission													
Proposal Presentation													
Check List submission													
Progress Presentation 1													
Final Check List submission													
Progress Presentation 2													
Final Report submission													
Website Submission													
Final Presentation and Viva													
Website evaluation & Logbook submission													
Research Paper submission													
Research Paper publication evidence submission													

Figure 9-1: Project Gantt chart and milestone timeline

10. REFERENCES LIST

The following references are formatted according to strict IEEE standards, prioritizing the scholarly sources that underpin the multiscale computer vision framework and generative synthesis methodology.

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[4] V. Mussenbrock, "face-api.js: JavaScript API for Face Detection and Face Recognition in the Browser with tensorflow.js," *GitHub*, 2020. [Online]. Available: https://github.com/justadudewhohacks/face-api.js. (Official source for the core framework used in the group project).

[5] J. Buolamwini and T. Gebru, "Gender Shades: Intersectional Accuracy Disparities in Commercial Gender Classification," in *Proc. Conference on Fairness, Accountability and Transparency*, 2018, pp. 77-91. (Critical reference for addressing **Demographic and Racial Bias** in facial analysis).

[6] Y. J. Heo, "Deepfake Detection using Vision Transformer," *arXiv preprint arXiv:2103.05610*, 2021. [Online]. Available: https://arxiv.org/abs/2103.05610. (Technical study on applying **ViT** for synthetic media detection).

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[12] W. Shao *et al.*, "TransForensics: Image forgery localization with dense self-attention," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 10, pp. 6489-6501, Oct. 2022

[13] K. Rossler *et al.*, "FaceForensics++: Learning to detect manipulated facial images," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1-11

[14] S. Agarwal *et al.*, "Protecting world leaders against deepfakes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR) Workshops*, June 2019, pp. 38-45

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11. APPENDICES

- ## Appendix A: Survey Instrument (Requirement Gathering).

To align the **"Digital Social Service"** with the practical needs of the Sri Lankan public, the following survey identifies user awareness and diagnostic requirements for deepfake detection.

- **Sample Questions for the General Public:**
  1. How often do you encounter images or videos on social media that you suspect are AI-generated (Deepfakes)?
  2. Are you aware of the impact of "Hyperrealism" where AI faces look more authentic than real ones?
  3. Would you trust a verification tool that explains *why* a photo is flagged (e.g., skin texture or landmark misalignment)?
  4. Is your primary device for browsing social media a high-end desktop or a mid-range smartphone?
  5. How critical is it for you that a detection tool works fairly across all **Sri Lankan cultural demographics**?

- ## Appendix B: Detailed Technical Diagrams

The following diagrams represent the internal logic and neural architecture of the proposed detection artifacts.

- **Vision Transformer (ViT) Patch Processing:** A flowchart demonstrating the division of a Sri Lankan facial image into 16×16 patches, followed by the self-attention mechanism to identify global structural inconsistencies.
- **StyleGAN2-ADA Feedback Loop:** A diagram showing the Adaptive Discriminator Augmentation process used to stabilize training on limited Sri Lankan datasets.
- **Face-API.js Landmark Integration:** A system flow diagram showing the interaction between the React-based frontend and the local TFLite model for real-time inference.

- ## Appendix C: Preliminary UI Mockups

The interface is designed as a **"Digital Social Service"** portal, prioritizing accessibility and transparency.

1. **The Public Verification Portal (Mobile View):**
   - **Scan/Upload Screen:** A simple drag-and-drop interface for users to submit suspected digital content.
   - **Diagnostic Report:** A result screen showing a "Authenticity Score" and an **Explainable AI (XAI)** heatmap highlighting detected anomalies in facial landmarks or skin textures.

2. **The Forensic Dashboard (Researcher View):**
    - **Metrics Screen:** Real-time monitoring of mAP and F1-scores across different generative techniques (GAN vs. Diffusion).
    - **Demographic Fairness Map:** A visualization tool to ensure the model maintains high accuracy across all local ethnicities.

Appendix D: Risk Analysis and Mitigation Matrix

The research identifies technical and ethical risks specifically related to AI-based facial analysis in a local context.

Risk Category	Identified Risk	Mitigation Strategy
Technical	Visual Mimicry: High-fidelity deepfakes bypassing standard CNN filters.	Implementing ViT and Swin-Base to capture long-range structural dependencies and micro-artifacts.
Data	Demographic Bias: Models performing poorly on specific Sri Lankan features.	Utilizing StyleGAN2-ADA to synthesize a demographically balanced dataset for training.
Operational	High Latency: Transformer models being too "heavy" for mobile browsers.	Applying INT8 Post-Training Quantization (PTQ) to reduce model size by 75% for sub-50ms inference.
Ethical	Dual-Use: The synthesis tool being used to create deepfakes instead of detecting them.	Maintaining all generative synthesis within a strictly controlled, offline research environment.

Appendix E: Ethical Clearance

As documented in the Topic Assessment Form (TAF-R26-IT-011), the research adheres to the following ethical protocols:

- **Anonymization:** All authentic facial data from Sri Lankan volunteers will be anonymized to protect personal identities during the training of ViT and Swin-Base models.
- **Informed Consent:** Field data collection will only occur after obtaining explicit permission from participants, ensuring they understand their data is used solely for deepfake detection research.
- **Compliance:** The project maintains compliance with the **National Digital Policy** and the ethical guidelines provided by SLIIT's research committee